BRAC University

Department of Mathematics and Natural Sciences

LECTURE ON

Real Analysis II (MAT321)

Path Connectedness

Connected vs Path Connected, Topologist Sine Curve, Local Connectedness

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CONDUCTED BY

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Motivation

Paths in Metric Spaces

Path

Let X be a metric space and let $a, b \in X$. A *path from a to b in X* is a continuous function

$$\gamma : [0,1] \to X$$

such that

$$\gamma(0) = a \qquad \text{and} \qquad \gamma(1) = b.$$

The points a and b are called the *initial point* and *terminal point* of the path.

Remark

The interval $[0,1]$ is used only as a convenient parameter interval. A path is not just the image $\gamma([0,1])$; it is the continuous parametrization γ together with the direction from a to b.

Straight-line path in $\mathbb{R}^n$

Let $a, b \in \mathbb{R}^n$. Define

$$\gamma(t) = (1-t)a + tb, \qquad 0 \leq t \leq 1.$$

Then γ is continuous, $\gamma(0) = a$, and $\gamma(1) = b$. Hence γ is a path from a to b in $\mathbb{R}^n$.

Path Connected Sets

Path Connected Set

Let X be a metric space. A subset $A \subseteq X$ is said to be *path connected* if for every pair of points $a, b \in A$, there exists a continuous function

$$\gamma : [0, 1] \to A$$

such that

$$\gamma(0) = a, \qquad \gamma(1) = b.$$

Intervals are path connected

Every interval $I \subseteq \mathbb{R}$ is path connected. If $a, b \in I$, define

$$\gamma(t) = (1-t)a + tb.$$

Since I is an interval, every point between a and b belongs to I. Hence $\gamma([0,1]) \subseteq I$, and γ is a path from a to b inside I.

Convex subsets of $\mathbb{R}^n$

A subset $C \subseteq \mathbb{R}^n$ is called convex if

$$(1-t)x + ty \in C \qquad \text{for all } x, y \in C \text{ and } 0 \leq t \leq 1.$$

Every convex set is path connected by the straight-line path.

Generalized Intermediate Value Theorem in Convex Sets

Let X be a convex path-connected metric space, and let $f : X \to \mathbb{R}$ be a continuous function. Suppose that for some $x_1, x_2 \in X$ and $K \in \mathbb{R}$, $f(x_1) < k < f(x_2)$. Then there exists a point x on the line segment joining x_1 and x_2 such that

$$f(x) = k.$$

Path Connected Implies Connected

Path connected $\Rightarrow$ connected

Every path connected subset of a metric space is connected.

Converse?

The converse is false. There exist connected sets that are not path connected. The most important problem for us is the topologist sine curve.

A Useful Tool: Pasting Paths

Joining two paths

Let X be a metric space. Suppose $\gamma_1 : [0,1] \to X$ is a path from a to b and $\gamma_2 : [0,1] \to X$ is a path from b to c. Define

$$\gamma(t) = \begin{cases} \gamma_1(2t), & 0 \le t \le \frac{1}{2}, \\ \gamma_2(2t-1), & \frac{1}{2} \le t \le 1. \end{cases}$$

Then γ is a path from a to c.

Proof. Clearly $\gamma(0) = \gamma_1(0) = a$ and $\gamma(1) = \gamma_2(1) = c$. The only point at which continuity needs checking is $t = \frac{1}{2}$.

At $t = \frac{1}{2}$, the left-hand formula gives

$$\gamma_1(1) = b,$$

and the right-hand formula gives

$$\gamma_2(0) = b.$$

Thus the two pieces agree at $t = \frac{1}{2}$. By the pasting lemma for continuous functions, γ is continuous on $[0,1]$. Hence γ is a path from a to c. $\square$

Remark

This theorem says that if a can be joined to b, and b can be joined to c, then a can be joined to c.

Union of Path Connected Sets

Union of two path connected sets

Let A and B be path connected subsets of a metric space X. If

$$A \cap B \neq \emptyset,$$

then $A \cup B$ is path connected.

Proof. Choose a point $p \in A \cap B$. Let $x, y \in A \cup B$. We must show that x and y can be joined by a path in $A \cup B$.

If $x \in A$, then because A is path connected, there is a path in A from x to p. If $x \in B$, then because B is path connected, there is a path in B from x to p. In either case, there is a path in $A \cup B$ from x to p.

Similarly, there is a path in $A \cup B$ from p to y. Joining these two paths gives a path from x to y in $A \cup B$. Therefore $A \cup B$ is path connected.
$\square$

Arbitrary union of path connected sets

Let $\{A_\alpha\}_{\alpha \in I}$ be a family of path connected subsets of a metric space X. If

$$\bigcap_{\alpha \in I} A_\alpha \neq \emptyset,$$

then

$$\bigcup_{\alpha \in I} A_\alpha$$

is path connected.

Topologist Sine Curve

Topologist sine curve

Let

$$G = \left\{ \left(x, \sin \frac{1}{x} \right) : x > 0 \right\}$$

and

$$V = \{(0, y) : -1 \leq y \leq 1\}.$$

The set

$$T = G \cup V$$

is called the *topologist sine curve*.

The topologist sine curve is connected

The set $T = G \cup V$ is connected.

Connected but Not Path Connected

Topologist sine curve is not path connected

The topologist sine curve

$$T = \left\{ \left(x, \sin \frac{1}{x} \right) : x > 0 \right\} \cup \{(0, y) : -1 \leq y \leq 1\}$$

is connected but not path connected.

Path Components

Path component

Let X be a metric space and let $x \in X$. The *path component of* x is the set of all points of X that can be joined to x by a path in X:

$$P_x = \{y \in X : \text{there exists a path in } X \text{ from } x \text{ to } y\}.$$

A *path component of* X is any such set P_x.

Remark

A path component is the largest path connected piece of X that contains the point x.

Problem

For

$$X = (0,1) \cup (2,3),$$

the path components are $(0,1)$ and $(2,3)$.

Problem

For $X = \mathbb{R}^n$, there is only one path component, namely $\mathbb{R}^n$ itself.

Properties of Path Components

Path components partition the space

Let X be a metric space. Then the path components of X are either identical or pairwise disjoint, and their union is all of X.

Proof. Every point $x \in X$ belongs to its own path component P_x, because the constant path $\gamma(t) = x$ joins x to itself. Hence the union of all path components is X.

Now suppose $P_x \cap P_y \neq \emptyset$. Choose $z \in P_x \cap P_y$. Then there is a path from x to z and a path from y to z. Reversing the second path gives a path from z to y. Joining paths gives a path from x to y.

Now if $w \in P_x$, then w can be joined to x, and x can be joined to y. Hence w can be joined to y, so $w \in P_y$. Thus $P_x \subseteq P_y$. Similarly $P_y \subseteq P_x$. Therefore $P_x = P_y$.

So path components are either identical or disjoint. $\square$

Maximality

Each path component of X is a maximal path connected subset of X.

Connected Components vs Path Components

Connected component

A *connected component* of X is a maximal connected subset of X.

Path component

A *path component* of X is a maximal path connected subset of X.

Path components lie inside connected components

Every path component of a metric space X is contained in a connected component of X.

Remark

Connected components may be larger than path components. They are equal in many familiar spaces, but not in all metric spaces.

Comparing Components

$\mathbb{R}^n$

The space $\mathbb{R}^n$ is path connected, hence connected. Therefore it has exactly one connected component and exactly one path component:

$$\mathbb{R}^n.$$

$(0,1) \cup (2,3)$

The set

$$X = (0,1) \cup (2,3)$$

has two connected components and two path components:

$$(0,1), \qquad (2,3).$$

Here connected components and path components are the same.

Topologist sine curve

For the topologist sine curve

$$T = G \cup V,$$

where

$$G = \left\{\left(x, \sin\frac{1}{x}\right) : x > 0\right\}, \qquad V = \{(0, y) : -1 \leq y \leq 1\},$$

there is one connected component, namely T itself. But there are two path components:

$$G \qquad \text{and} \qquad V.$$

Thus connected components and path components can be different.

Totally Disconnected Spaces

Totally disconnected space

A metric space X is called *totally disconnected* if every connected component of X is a singleton set.

Remark

Totally disconnected does not mean that the space has no limit points. For problem, $\mathbb{Q}$ is totally disconnected, but every rational number is a limit point of $\mathbb{Q}$.

A finite metric space

Every finite metric space with the discrete metric is totally disconnected, because every singleton is open and closed.

$\mathbb{Q}$ is totally disconnected

The metric space $(\mathbb{Q}, d)$, with the usual metric inherited from $\mathbb{R}$, is totally disconnected. Hence the components of $\mathbb{Q}$ are precisely the singleton sets

$$\{q\}, \qquad q \in \mathbb{Q}.$$

Proof. Let C be a connected subset of $\mathbb{Q}$. We show that C has at most one point.

Suppose, for contradiction, that C contains two distinct rational numbers $p < q$. Since irrational numbers are dense in $\mathbb{R}$, choose an irrational number α such that

$$p < \alpha < q.$$

Define

$$U = C \cap (-\infty, \alpha), \qquad V = C \cap (\alpha, \infty).$$

Then U and V are non-empty, because $p \in U$ and $q \in V$. Also

$$C = U \cup V, \qquad U \cap V = \emptyset.$$

Moreover, U and V are separated in C, because the cut point α is not rational and hence does not belong to $\mathbb{Q}$. Thus C is disconnected, which is a contradiction.

Therefore every connected subset of $\mathbb{Q}$ contains at most one point. Hence every connected component of $\mathbb{Q}$ is a singleton. $\square$

The irrationals are totally disconnected

The metric space $\mathbb{R} \setminus \mathbb{Q}$, with the usual metric inherited from $\mathbb{R}$, is totally disconnected.

Proof. Let C be a connected subset of $\mathbb{R} \setminus \mathbb{Q}$. Suppose that C contains two distinct irrational numbers $a < b$.

Since rational numbers are dense in $\mathbb{R}$, choose a rational number r such that

$$a < r < b.$$

Define

$$U = C \cap (-\infty, r), \qquad V = C \cap (r, \infty).$$

Then U and V are non-empty, disjoint, and their union is C. Since $r \notin \mathbb{R} \setminus \mathbb{Q}$, the sets U and V form a separation of C. Therefore C is disconnected, a contradiction.

Thus every connected subset of $\mathbb{R} \setminus \mathbb{Q}$ has at most one point. Hence the components of $\mathbb{R} \setminus \mathbb{Q}$ are singleton sets. $\square$

Remark

Both $\mathbb{Q}$ and $\mathbb{R} \setminus \mathbb{Q}$ are dense in $\mathbb{R}$, and both are totally disconnected. Density and connectedness are very different ideas.

Totally Disconnected vs Discrete

Discrete metric space

A metric space X is called *discrete* if every singleton set $\{x\}$ is open in X.

Discrete implies totally disconnected

Every discrete metric space is totally disconnected.

Proof. Let X be discrete. Suppose $C \subseteq X$ contains two distinct points a and b. Since $\{a\}$ is open and closed in X, the set

$$C \cap \{a\} = \{a\}$$

is a non-empty proper clopen subset of C. Therefore C is disconnected. Hence no connected subset of X can contain more than one point. So X is totally disconnected. $\square$

Converse is false

A totally disconnected space need not be discrete. The space $\mathbb{Q}$ is totally disconnected, but it is not discrete under the usual metric. Indeed, every open ball around a rational number contains infinitely many rational numbers.

The Cantor Set

Cantor set

The Cantor set $\mathcal{C}$ is obtained from $[0, 1]$ by repeatedly removing open middle thirds:

$$[0,1] \supset \left[0, \frac{1}{3}\right] \cup \left[\frac{2}{3}, 1\right] \supset \cdots .$$

Equivalently,

$$\mathcal{C} = \bigcap_{n=0}^{\infty} C_n,$$

where C_n is the union of the closed intervals remaining after the n-th step.

Cantor set is totally disconnected

The Cantor set $\mathcal{C}$ is totally disconnected.

Proof Sketch. Let $x, y \in \mathcal{C}$ with $x < y$. During the construction of the Cantor set, some removed middle-third interval separates x from y. That is, there exists an open interval (a, b) removed at some finite stage such that

$$x < a < b < y.$$

Then

$$\mathcal{C} \cap (-\infty, a] \qquad \text{and} \qquad \mathcal{C} \cap [b, \infty)$$

separate x and y inside $\mathcal{C}$. Therefore no connected subset of $\mathcal{C}$ can contain two distinct points. Hence every component of $\mathcal{C}$ is a singleton. $\square$

Remark

The Cantor set is compact, uncountable, and has no isolated points, but it is still totally disconnected. This makes it one of the most important problems in real analysis and topology.

Local Connectedness

Locally path connected

A metric space X is called *locally path connected* if for every point $x \in X$ and every open set U containing x, there exists an open path connected set V such that

$$x \in V \subseteq U.$$

Locally connected

A metric space X is called *locally connected* if for every point $x \in X$ and every open set U containing x, there exists an open connected set V such that

$$x \in V \subseteq U.$$

Problem

Local connectedness describes the small neighborhoods around each point. It is possible for a space to be connected but fail to be locally connected.

Problem

Every open subset of $\mathbb{R}^n$ is locally path connected. Around any point, choose a sufficiently small open ball contained in the given open set. Open balls in $\mathbb{R}^n$ are path connected.

When Components and Path Components Agree

Locally path connected spaces

If X is locally path connected, then the path components of X are open. Moreover, in a locally path connected space, connected components and path components are the same.

Proof Sketch. Let P be a path component of X and let $x \in P$. Since X is locally path connected, there exists an open path connected neighborhood V of x. Since V is path connected and contains x, every point of V is path connected to x. Thus

$$V \subseteq P.$$

So every point of P has an open neighborhood contained in P. Hence P is open.

Now let C be a connected component of X. Since path components are open and partition X, if C met two different path components, then those open path components would give a separation of C. This is impossible because C is connected. Hence C is contained in one path component. But every path component is connected, so maximality gives equality. $\square$

Problem

This theorem explains why in familiar spaces such as open subsets of $\mathbb{R}^n$, connected components and path components usually coincide.

Thank You!

We'd love your questions and feedback.

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References

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